HDOC DAY-63

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BATCH-15

AIM:

To write a C program to find the missing number in an integer array using pointers, functions, and a Variable Length Array (VLA).

QUESTION:

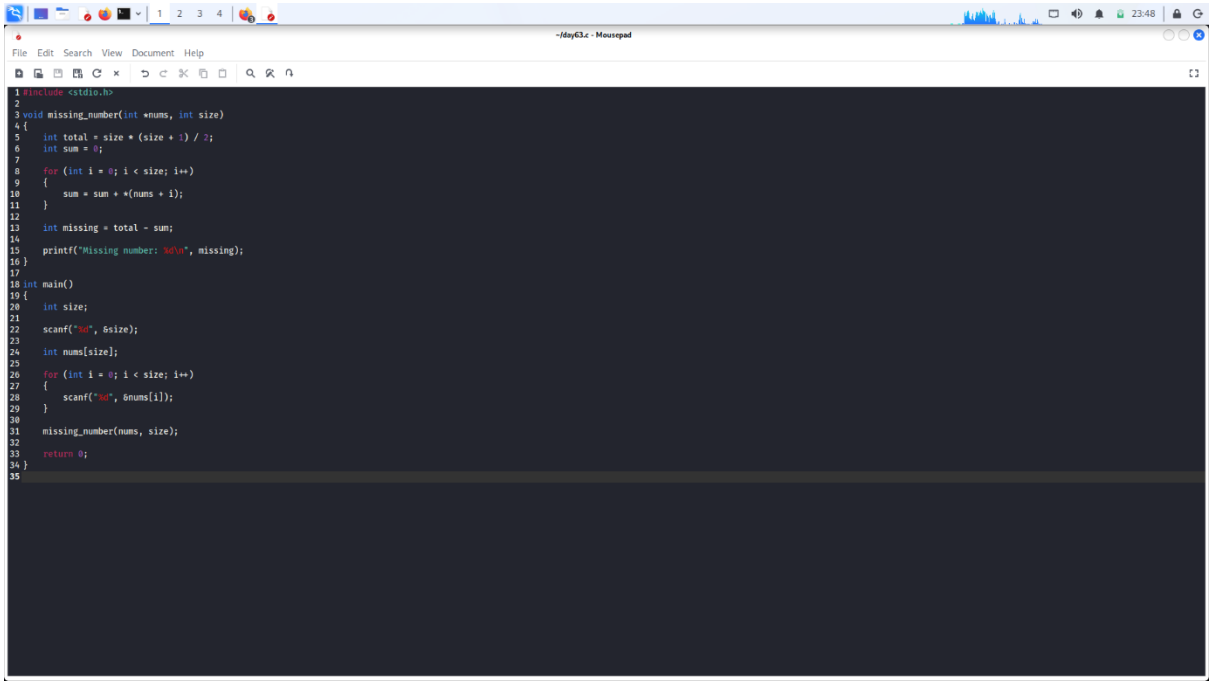
Write a C program that reads the size and elements of a variable-length integer array. The array contains distinct integers from 0 to size, with exactly one number missing. Pass the array to a function named missing_number() using pointers and display the missing integer.

ALGORITHM:

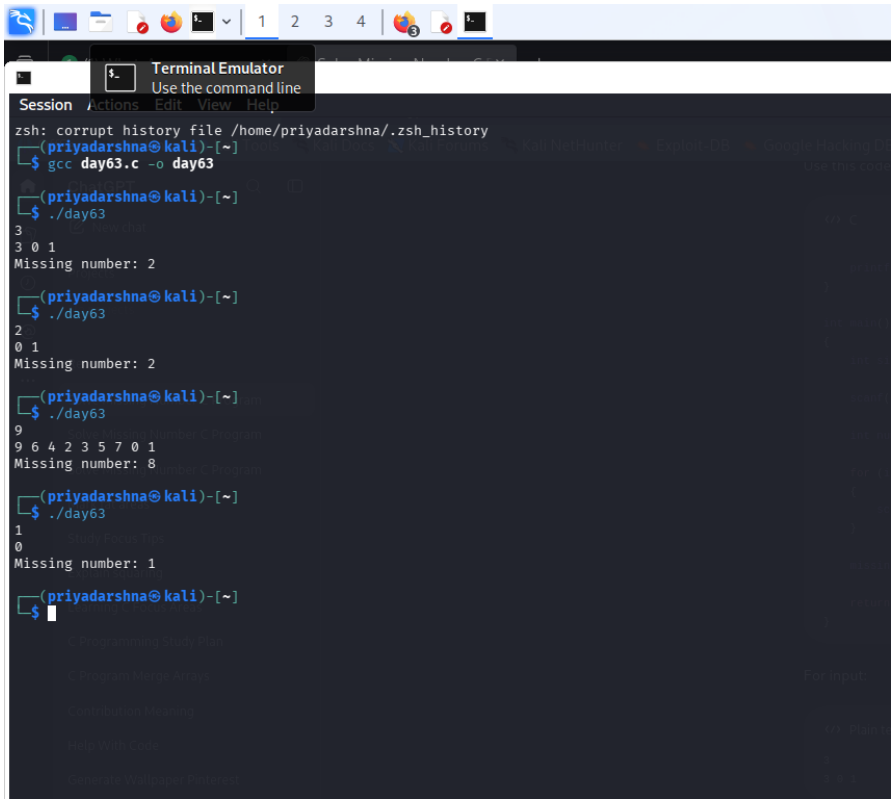
1. Start the program.
2. Read the size of the array.
3. Declare a variable-length array nums[size].
4. Read all the elements of the array.
5. Pass the array and size to the missing_number() function using a pointer.
6. Calculate the expected sum using:
   total = size * (size + 1) / 2
7. Calculate the sum of all elements present in the array.
8. Find the missing number using:
   missing = total - sum
9. Display the missing number.

10. Stop the program.

>C PROGRAM:



>OUTPUT


Terminal Emulator
Use the command line
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna㉿kali)-[~]
$ gcc day63.c -o day63
(priyadarshna㉿kali)-[~]
$ ./day63
3
3 0 1
Missing number: 2
(priyadarshna㉿kali)-[~]
$ ./day63
2
0 1
Missing number: 2
(priyadarshna㉿kali)-[~]
$ ./day63
9
9 6 4 2 3 5 7 0 1
Missing number: 8
(priyadarshna㉿kali)-[~]
$ ./day63
1
0
Missing number: 1
(priyadarshna㉿kali)-[~]
$